SUPERMARKET SALES DATA ANALYSIS REPORT

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PROJECT OUTLINE :

This report presents a comprehensive analysis of supermarket sales data to identify trends in product performance, customer purchasing behaviour, and payment methods. The goal is to provide actionable business insights that can support better retail decision-making.

PROJECT SUMMARY :

The purpose of this project is to analyse supermarket transaction data and uncover patterns in sales performance, customer preferences, and payment behaviours. Using Python-based data analysis tools, the dataset was cleaned, processed, and explored to generate meaningful insights. The dataset contains 2000 transaction records including product line information, quantities purchased, branch locations, customer types, payment methods, and customer ratings. Exploratory data analysis revealed that certain product categories generate higher revenue and that sales vary between supermarket branches. Additionally, digital payment methods such as card and e-wallets are widely used by customers. The analysis also indicates that customer satisfaction ratings are generally high. These insights can help supermarket managers improve inventory planning, optimize marketing strategies, and enhance customer experience. Overall, this analysis demonstrates the value of data analytics in supporting data-driven retail management decisions.

METHODOLOGY :

The analysis followed a structured data analytics workflow to ensure reliable results and meaningful insights.

STEP	DESCRIPTION
Data Loading	Dataset imported using pandas from CSV format.
Data Cleaning	Checked for missing values, duplicates, and incorrect data types.
Data Transformation	Converted date and time fields and created new features such as month and day.
Exploratory Analysis	Analyzed product sales, branch performance, and customer behavior.
Visualization	Created charts including bar charts, line charts, and heatmaps using matplotlib and seaborn.
Insight Generation	Interpreted patterns and generated business recommendations.

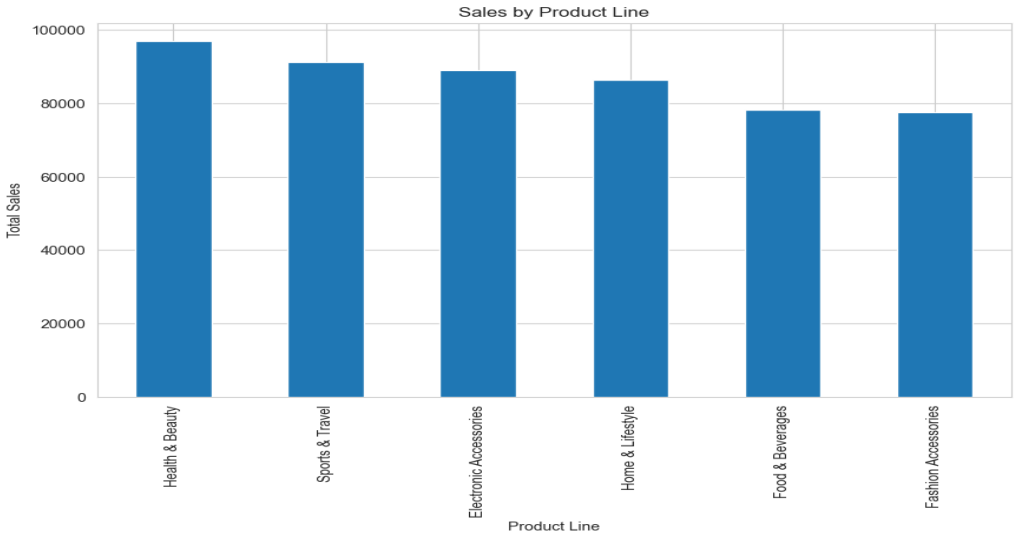
TOOLS USED :

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook were used for performing data cleaning, analysis, and visualization.

ANALYSIS FINDINGS (KEY ANALYSIS AREAS) :

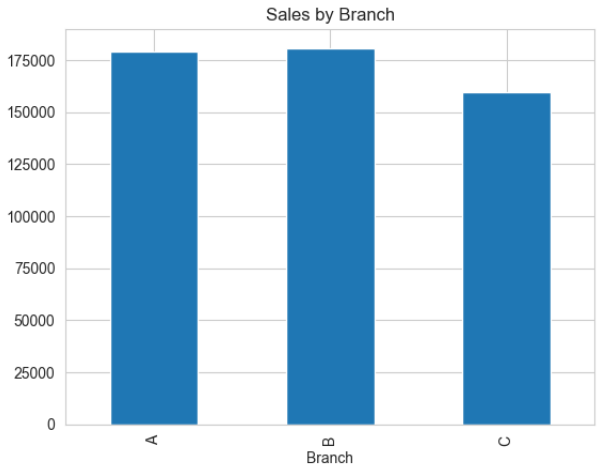
- **Product Line Performance :**

  Analysis shows that some product lines generate significantly higher revenue compared to others. These categories represent the core revenue drivers for the supermarket.



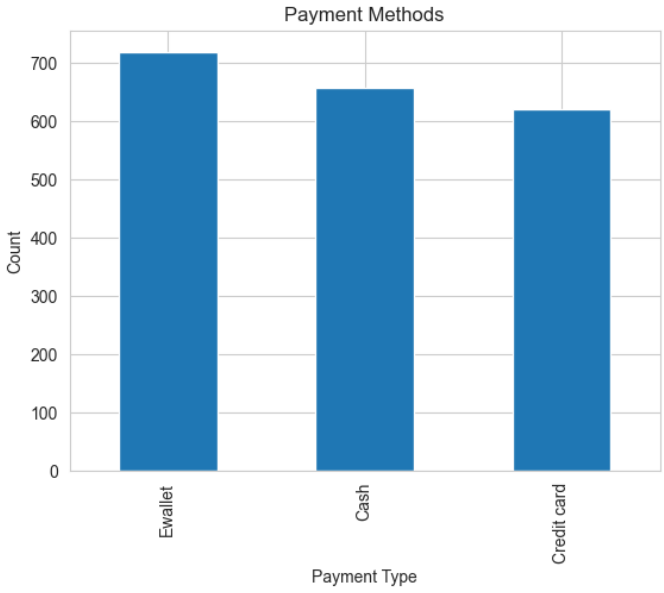
- **Branch Sales Comparison :**

  Sales vary between different branches, suggesting that location, customer demographics, and store size may influence revenue generation.



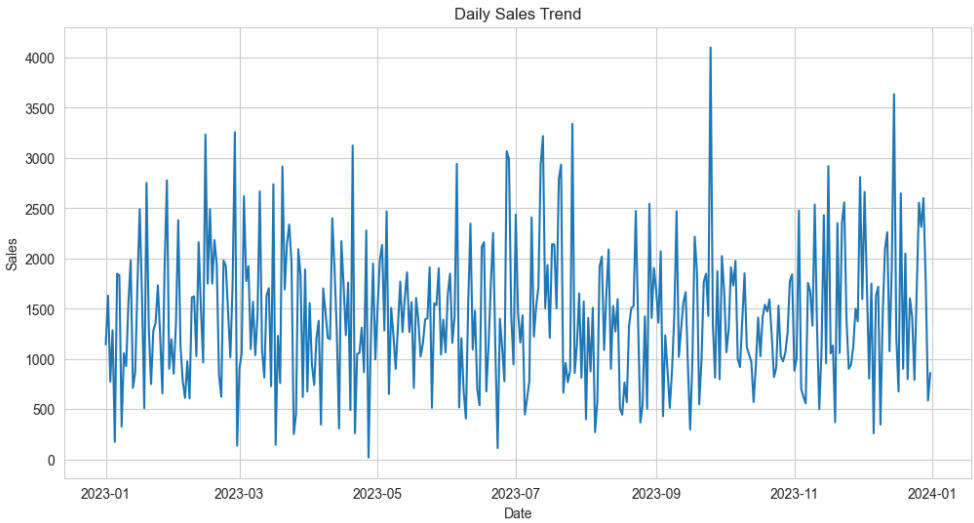
- **Payment method Distribution :**

The majority of transactions are conducted using digital payment methods such as cards and e-wallets, indicating a shift towards cashless transactions.



- **Daily Sales Trend :**

Sales trends fluctuate over time with certain periods showing increased demand. This information can help in planning inventory and staffing during peak periods.



STATISTICAL INSIGHTS :

Statistical analysis was conducted on numerical variables such as unit price, quantity, tax, total sales value, and customer ratings. Summary statistics including mean, minimum, maximum, and standard deviation were used to understand the distribution of values. Correlation analysis revealed relationships between variables such as quantity purchased and total sales value. A strong positive relationship indicates that larger purchase quantities directly contribute to higher transaction totals.

KEY FINDINGS :

- Certain product lines contribute significantly higher revenue.
- Sales distribution differs across supermarket branches.
- Customer type influences purchasing patterns.
- Digital payment methods are commonly used.
- Customer ratings indicate generally positive satisfaction levels.

BUSINESS RECOMMENDATIONS :

- Increase inventory levels for high-performing product lines.
- Introduce promotional campaigns for lower-performing categories.
- Encourage digital payment usage by offering loyalty rewards or discounts.
- Allocate more resources, staff and marketing during peak sales periods.
- Maintain high service standards to preserve positive customer ratings.

CONCLUSION :

This project demonstrates how data analytics can support strategic decision-making in retail environments. By analyzing supermarket sales data, we identified patterns in product performance, customer purchasing behavior, and payment preferences. These insights allow businesses to optimize inventory management, improve marketing strategies, and enhance customer experience. Future work could include predictive analytics and machine learning models to forecast sales and better anticipate customer demand.